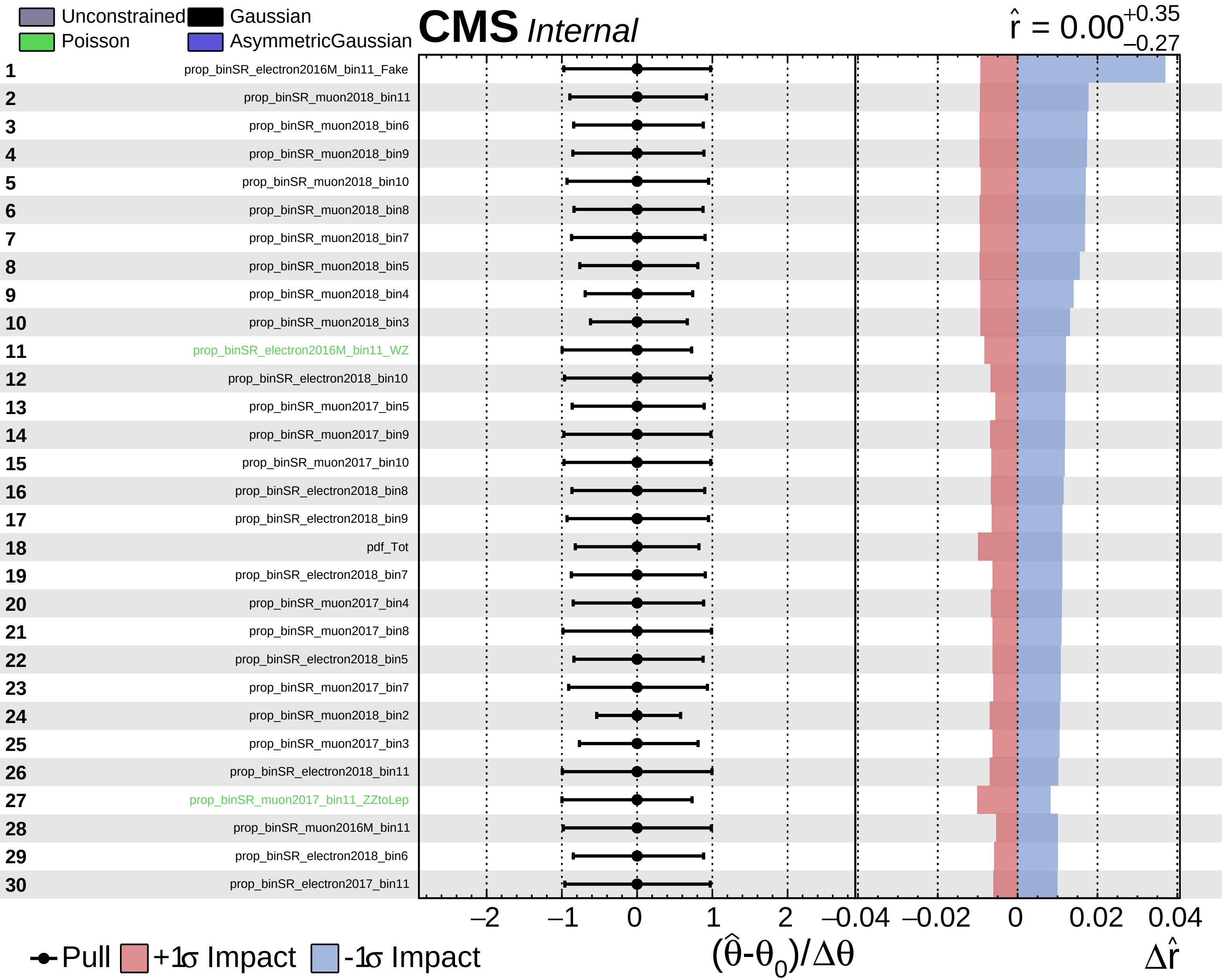
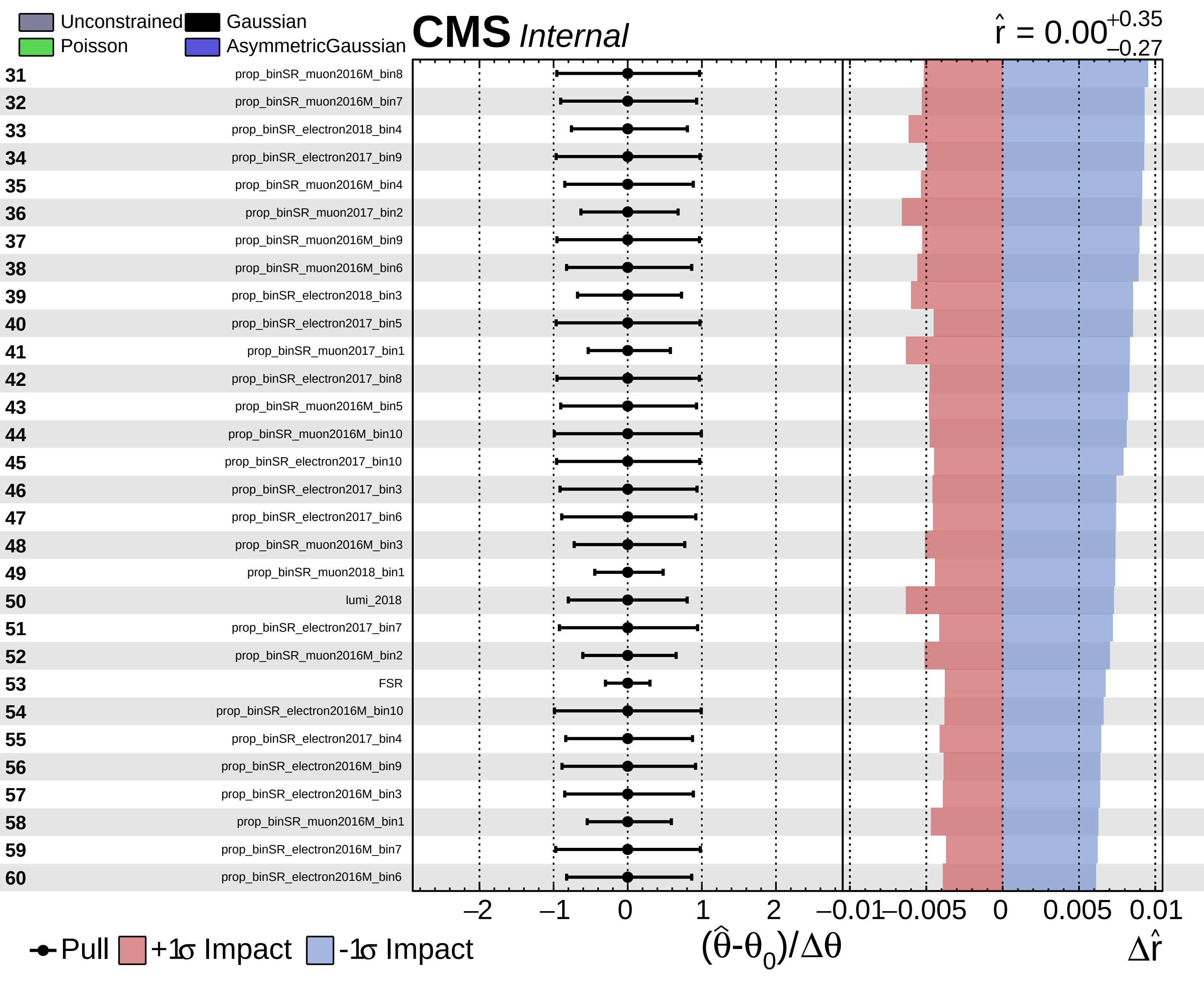
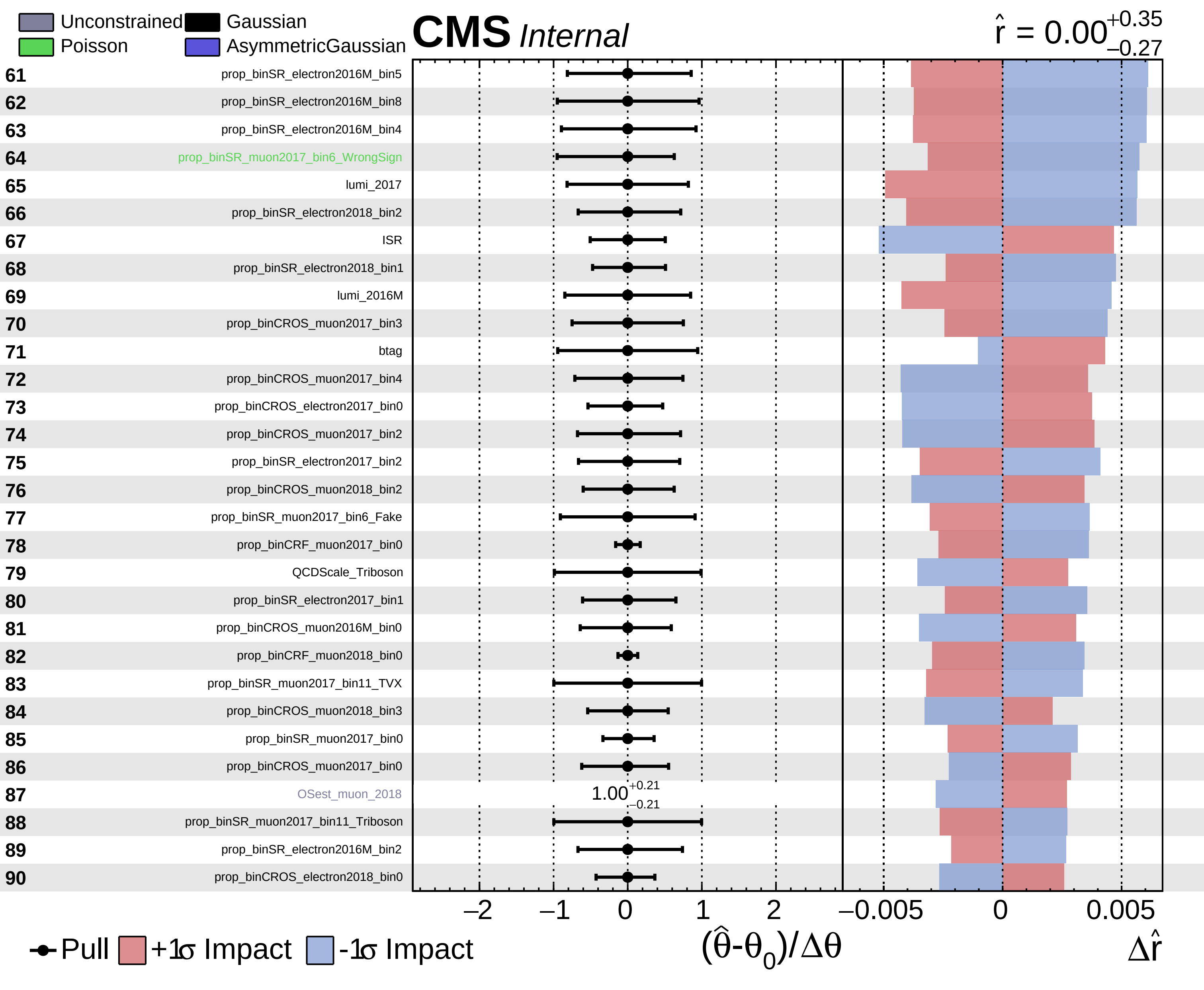
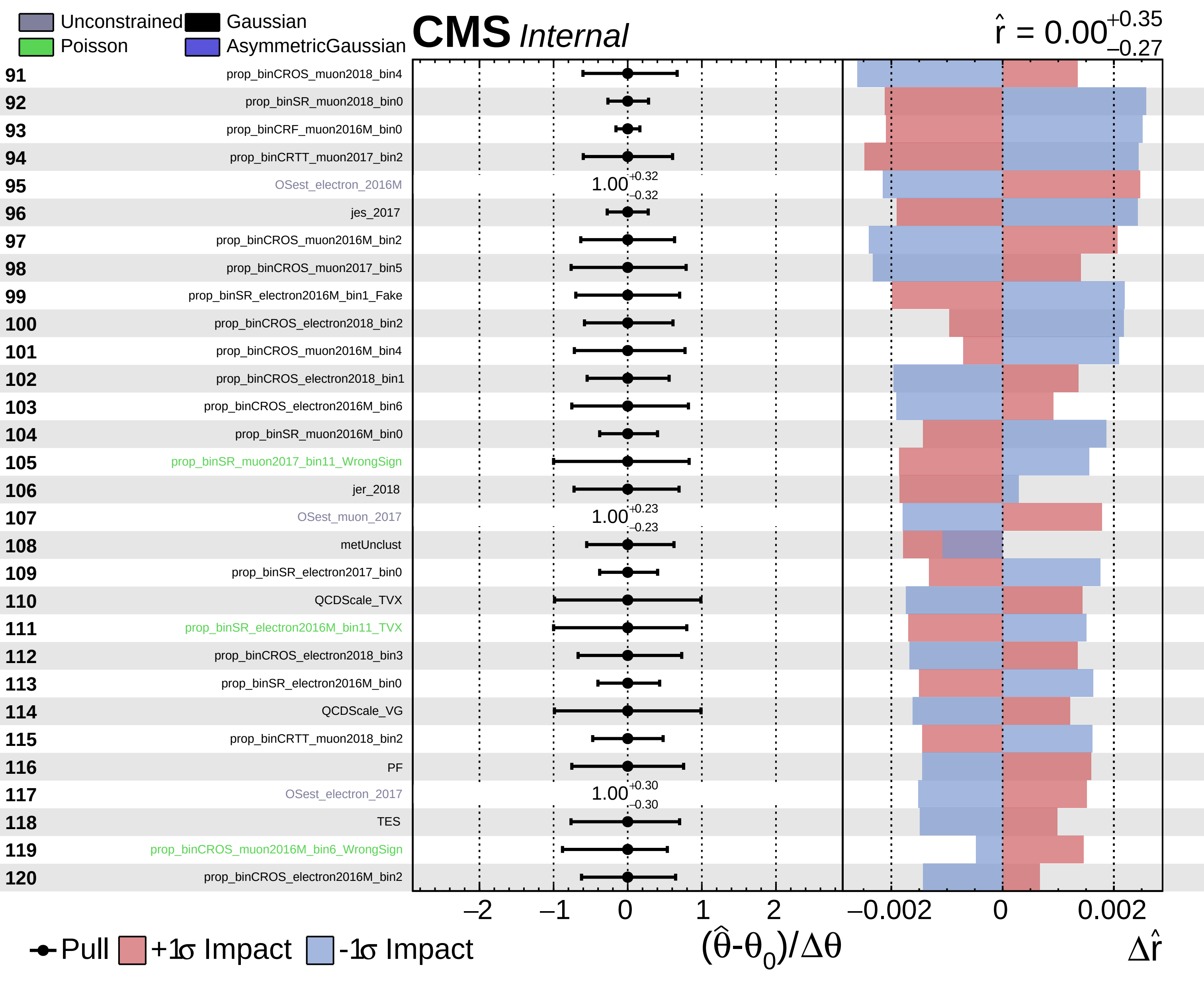
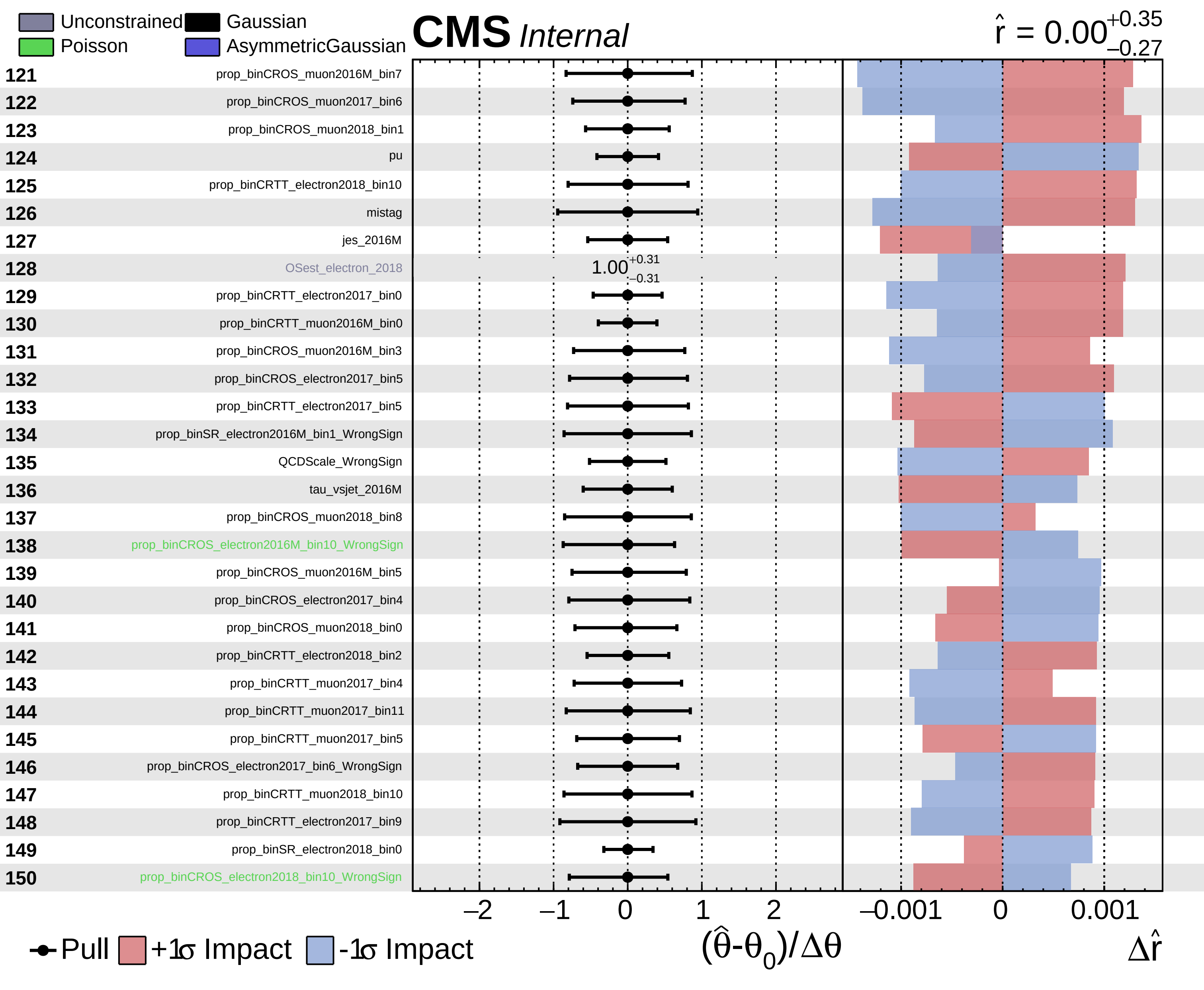

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.27}$
1 prop_binSR_electron2016M_bin11_Fake
2 prop_binSR_muon2018_bin11
3 prop_binSR_muon2018_bin6
4 prop_binSR_muon2018_bin9
5 prop_binSR_muon2018_bin10
6 prop_binSR_muon2018_bin8
7 prop_binSR_muon2018_bin7
8 prop_binSR_muon2018_bin5
9 prop_binSR_muon2018_bin4
10 prop_binSR_muon2018_bin3
11 prop_binSR_electron2016M_bin11_WZ
12 prop_binSR_electron2018_bin10
13 prop_binSR_muon2017_bin5
14 prop_binSR_muon2017_bin9
15 prop_binSR_muon2017_bin10
16 prop_binSR_electron2018_bin8
17 prop_binSR_electron2018_bin9
18 pdf_Tot
19 prop_binSR_electron2018_bin7
20 prop_binSR_muon2017_bin4
21 prop_binSR_muon2017_bin8
22 prop_binSR_electron2018_bin5
23 prop_binSR_muon2017_bin7
24 prop_binSR_muon2018_bin2
25 prop_binSR_muon2017_bin3
26 prop_binSR_electron2018_bin11
27 prop_binSR_muon2017_bin11_ZZtoLep
28 prop_binSR_muon2016M_bin11
29 prop_binSR_electron2018_bin6
30 prop_binSR_electron2017_bin11
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.27}$
31 prop_binSR_muon2016M_bin8
32 prop_binSR_muon2016M_bin7
33 prop_binSR_electron2018_bin4
34 prop_binSR_electron2017_bin9
35 prop_binSR_muon2016M_bin4
36 prop_binSR_muon2017_bin2
37 prop_binSR_muon2016M_bin9
38 prop_binSR_muon2016M_bin6
39 prop_binSR_electron2018_bin3
40 prop_binSR_electron2017_bin5
41 prop_binSR_muon2017_bin1
42 prop_binSR_electron2017_bin8
43 prop_binSR_muon2016M_bin5
44 prop_binSR_muon2016M_bin10
45 prop_binSR_electron2017_bin10
46 prop_binSR_electron2017_bin3
47 prop_binSR_electron2017_bin6
48 prop_binSR_muon2016M_bin3
49 prop_binSR_muon2018_bin1
50 lumi_2018
51 prop_binSR_electron2017_bin7
52 prop_binSR_muon2016M_bin2
53 FSR
54 prop_binSR_electron2016M_bin10
55 prop_binSR_electron2017_bin4
56 prop_binSR_electron2016M_bin9
57 prop_binSR_electron2016M_bin3
58 prop_binSR_muon2016M_bin1
59 prop_binSR_electron2016M_bin7
60 prop_binSR_electron2016M_bin6
−2
−1
0
1
2
−0.01
−0.005
0
0.005
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

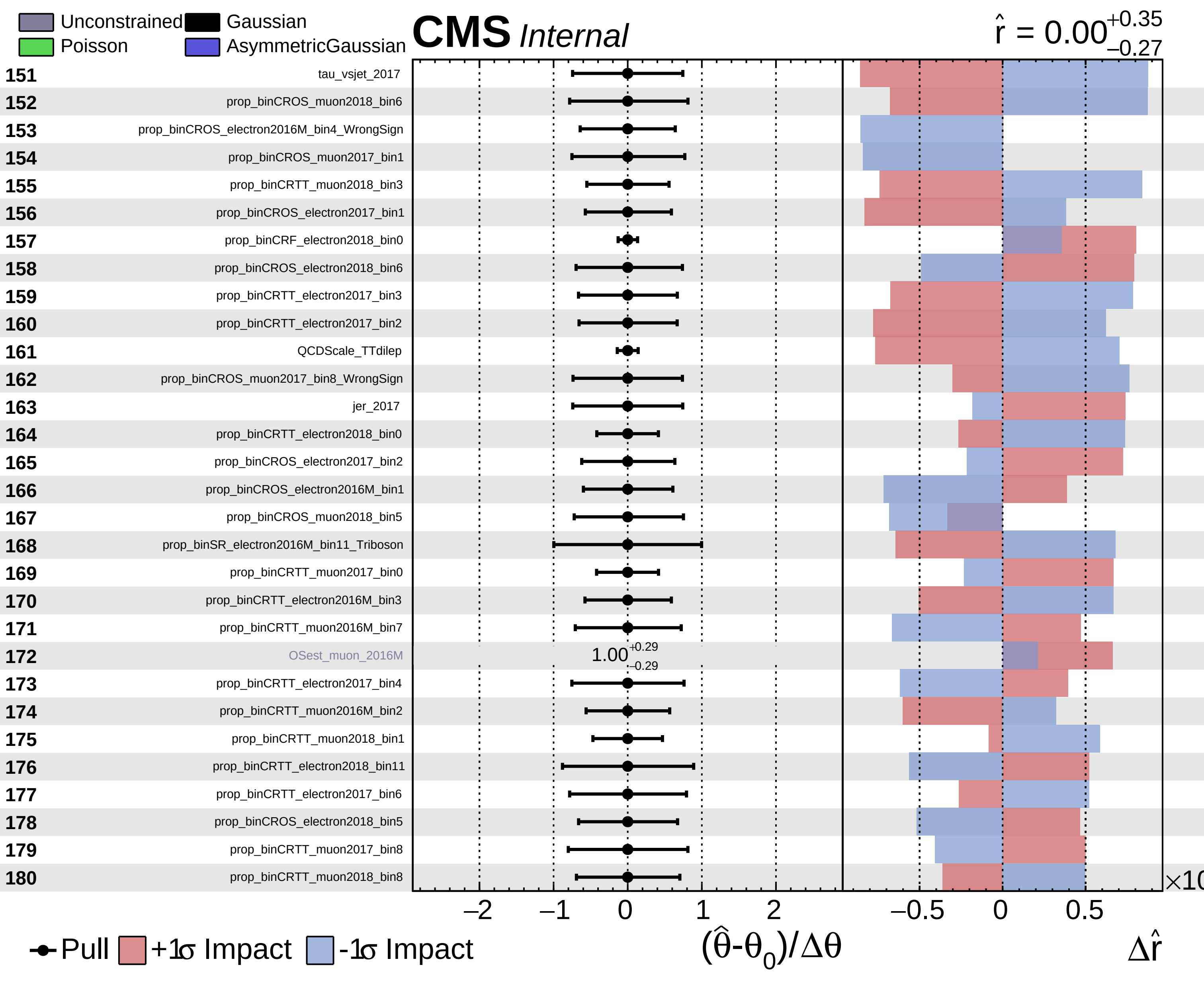


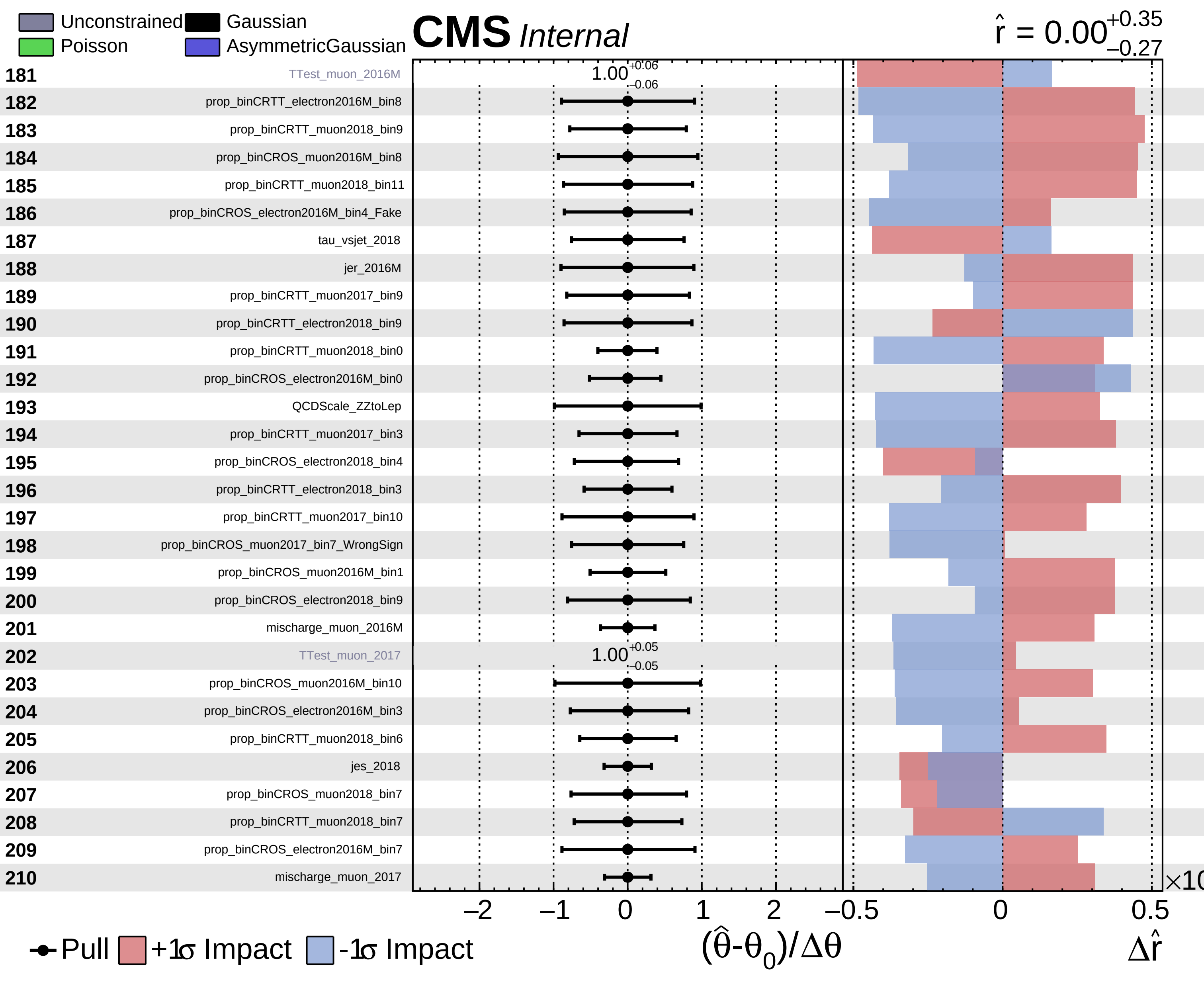
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.27}$
61 prop_binSR_electron2016M_bin5
62 prop_binSR_electron2016M_bin8
63 prop_binSR_electron2016M_bin4
64 prop_binSR_muon2017_bin6_WrongSign
65 lumi_2017
66 prop_binSR_electron2018_bin2
67 ISR
68 prop_binSR_electron2018_bin1
69 lumi_2016M
70 prop_binCROS_muon2017_bin3
71 btag
72 prop_binCROS_muon2017_bin4
73 prop_binCROS_electron2017_bin0
74 prop_binCROS_muon2017_bin2
75 prop_binSR_electron2017_bin2
76 prop_binCROS_muon2018_bin2
77 prop_binSR_muon2017_bin6_Fake
78 prop_binCRF_muon2017_bin0
79 QCDScale_Triboson
80 prop_binSR_electron2017_bin1
81 prop_binCROS_muon2016M_bin0
82 prop_binCRF_muon2018_bin0
83 prop_binSR_muon2017_bin11_TVX
84 prop_binCROS_muon2018_bin3
85 prop_binSR_muon2017_bin0
86 prop_binCROS_muon2017_bin0
87 OSest_muon_2018
$1.00^{+0.21}_{-0.21}$
88 prop_binSR_muon2017_bin11_Triboson
89 prop_binSR_electron2016M_bin2
90 prop_binCROS_electron2018_bin0
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.35 −0.27
91 prop_binCROS_muon2018_bin4
92 prop_binSR_muon2018_bin0
93 prop_binCRF_muon2016M_bin0
94 prop_binCRTT_muon2017_bin2
95 OSest_electron_2016M
1.00 +0.32 −0.32
96 jes_2017
97 prop_binCROS_muon2016M_bin2
98 prop_binCROS_muon2017_bin5
99 prop_binSR_electron2016M_bin1_Fake
100 prop_binCROS_electron2018_bin2
101 prop_binCROS_muon2016M_bin4
102 prop_binCROS_electron2018_bin1
103 prop_binCROS_electron2016M_bin6
104 prop_binSR_muon2016M_bin0
105 prop_binSR_muon2017_bin11_WrongSign
106 jer_2018
107 OSest_muon_2017
1.00 +0.23 −0.23
108 metUnclust
109 prop_binSR_electron2017_bin0
110 QCDScale_TVX
111 prop_binSR_electron2016M_bin11_TVX
112 prop_binCROS_electron2018_bin3
113 prop_binSR_electron2016M_bin0
114 QCDScale_VG
115 prop_binCRTT_muon2018_bin2
116 PF
117 OSest_electron_2017
1.00 +0.30 −0.30
118 TES
119 prop_binCROS_muon2016M_bin6_WrongSign
120 prop_binCROS_electron2016M_bin2
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

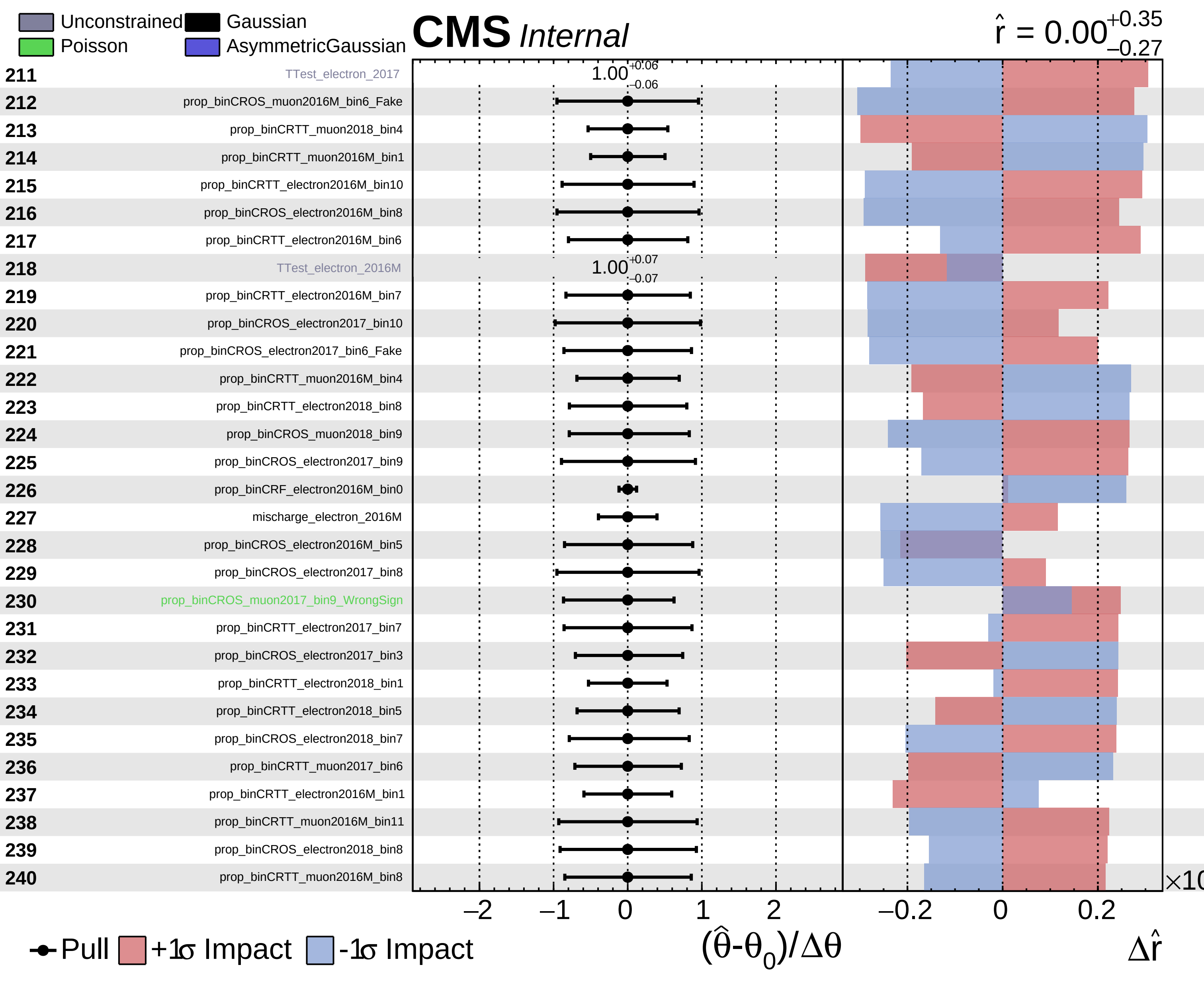


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.27}$
121 prop_binCROS_muon2016M_bin7
122 prop_binCROS_muon2017_bin6
123 prop_binCROS_muon2018_bin1
124 pu
125 prop_binCRTT_electron2018_bin10
126 mistag
127 jes_2016M
128 OSest_electron_2018
$1.00^{+0.31}_{-0.31}$
129 prop_binCRTT_electron2017_bin0
130 prop_binCRTT_muon2016M_bin0
131 prop_binCROS_muon2016M_bin3
132 prop_binCROS_electron2017_bin5
133 prop_binCRTT_electron2017_bin5
134 prop_binSR_electron2016M_bin1_WrongSign
135 QCDScale_WrongSign
136 tau_vsjet_2016M
137 prop_binCROS_muon2018_bin8
138 prop_binCROS_electron2016M_bin10_WrongSign
139 prop_binCROS_muon2016M_bin5
140 prop_binCROS_electron2017_bin4
141 prop_binCROS_muon2018_bin0
142 prop_binCRTT_electron2018_bin2
143 prop_binCRTT_muon2017_bin4
144 prop_binCRTT_muon2017_bin11
145 prop_binCRTT_muon2017_bin5
146 prop_binCROS_electron2017_bin6_WrongSign
147 prop_binCRTT_muon2018_bin10
148 prop_binCRTT_electron2017_bin9
149 prop_binSR_electron2018_bin0
150 prop_binCROS_electron2018_bin10_WrongSign
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

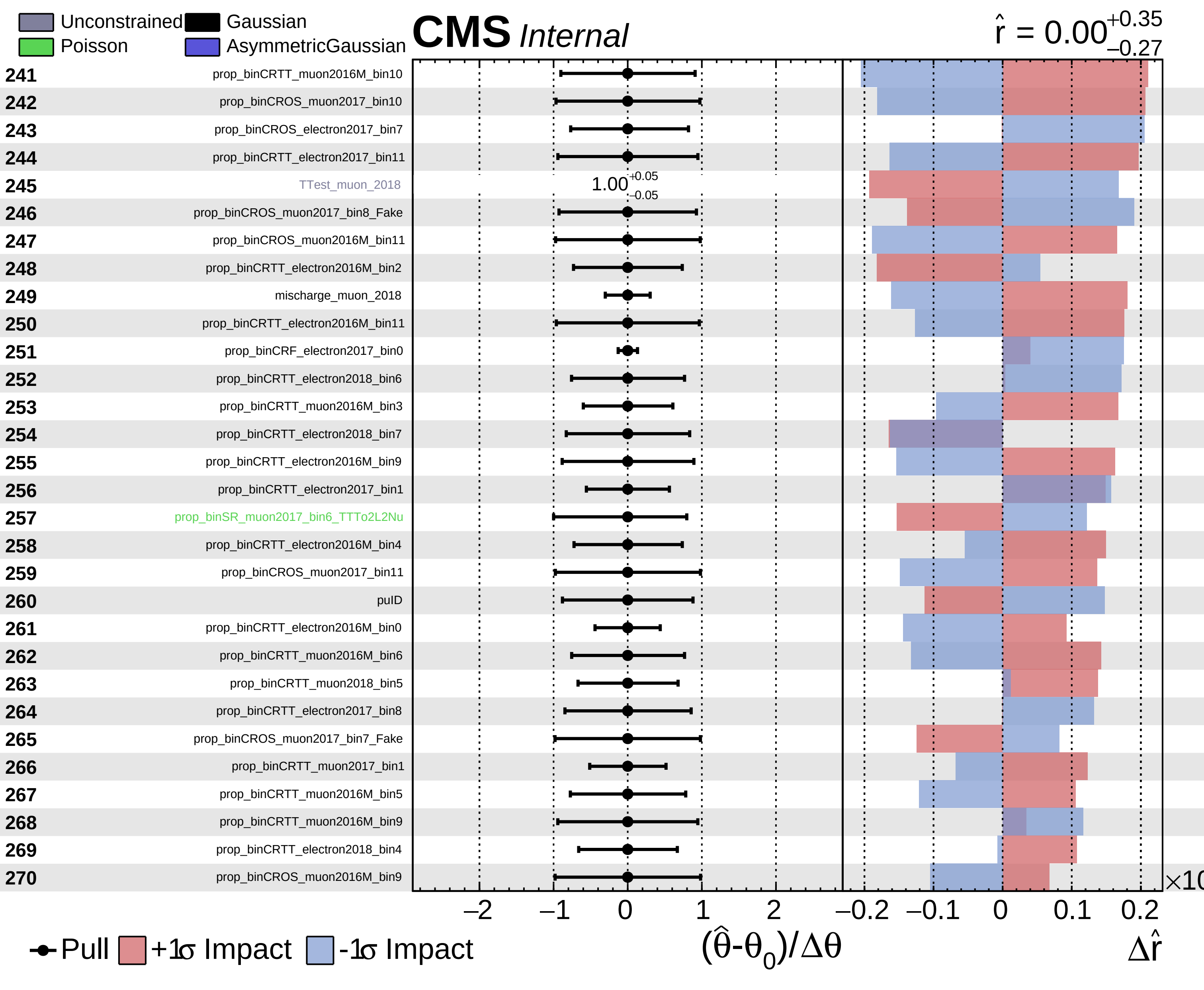

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.35 −0.27
151 tau_vsjet_2017
152 prop_binCROS_muon2018_bin6
153 prop_binCROS_electron2016M_bin4_WrongSign
154 prop_binCROS_muon2017_bin1
155 prop_binCRTT_muon2018_bin3
156 prop_binCROS_electron2017_bin1
157 prop_binCRF_electron2018_bin0
158 prop_binCROS_electron2018_bin6
159 prop_binCRTT_electron2017_bin3
160 prop_binCRTT_electron2017_bin2
161 QCDScale_TTdilep
162 prop_binCROS_muon2017_bin8_WrongSign
163 jer_2017
164 prop_binCRTT_electron2018_bin0
165 prop_binCROS_electron2017_bin2
166 prop_binCROS_electron2016M_bin1
167 prop_binCROS_muon2018_bin5
168 prop_binSR_electron2016M_bin11_Triboson
169 prop_binCRTT_muon2017_bin0
170 prop_binCRTT_electron2016M_bin3
171 prop_binCRTT_muon2016M_bin7
172 OSest_muon_2016M
1.00 +0.29 −0.29
173 prop_binCRTT_electron2017_bin4
174 prop_binCRTT_muon2016M_bin2
175 prop_binCRTT_muon2018_bin1
176 prop_binCRTT_electron2018_bin11
177 prop_binCRTT_electron2017_bin6
178 prop_binCROS_electron2018_bin5
179 prop_binCRTT_muon2017_bin8
180 prop_binCRTT_muon2018_bin8
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



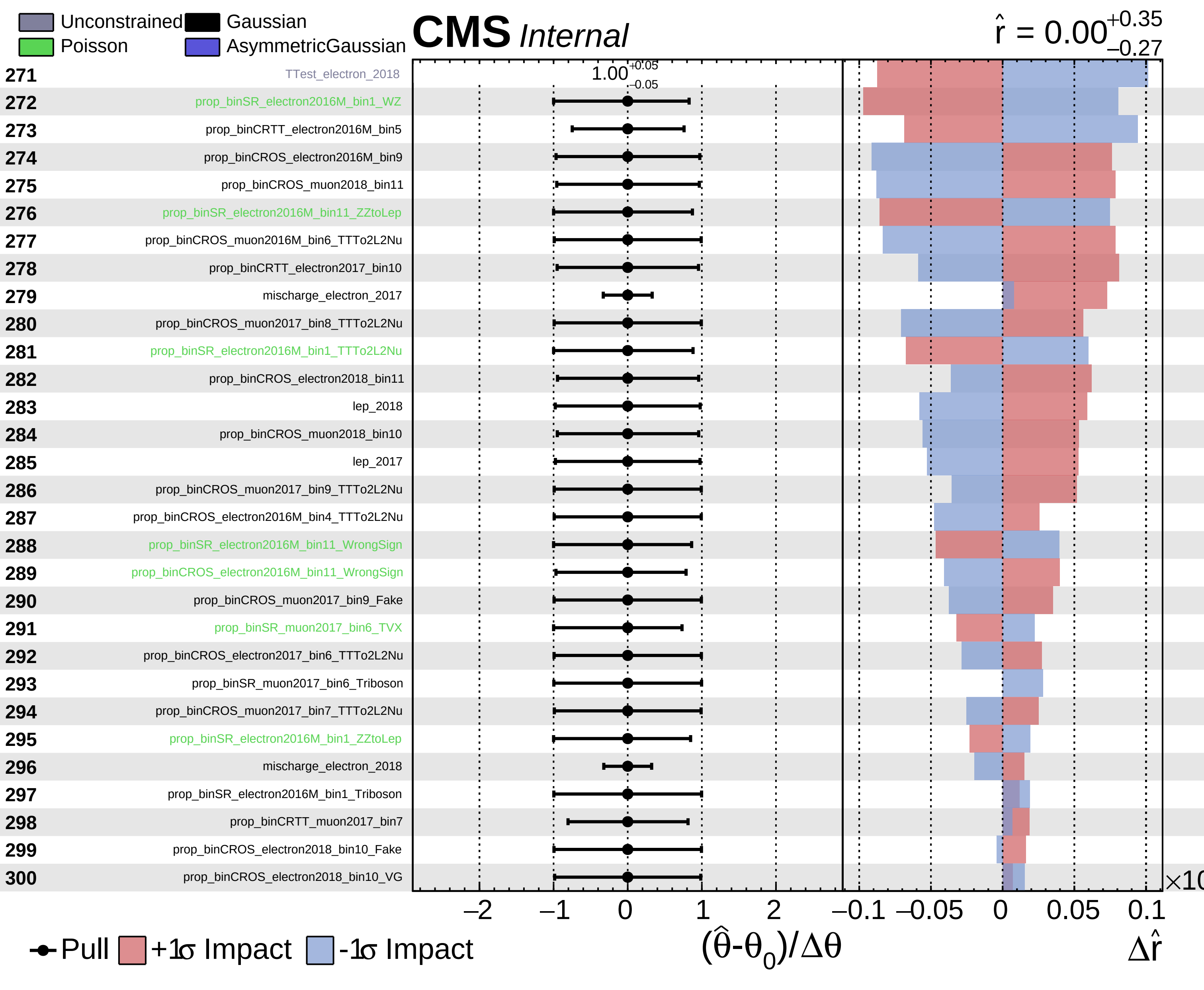
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.27}$
181 TTest_muon_2016M $1.00^{+0.06}_{-0.06}$
182 prop_binCRTT_electron2016M_bin8
183 prop_binCRTT_muon2018_bin9
184 prop_binCROS_muon2016M_bin8
185 prop_binCRTT_muon2018_bin11
186 prop_binCROS_electron2016M_bin4_Fake
187 tau_vsjet_2018
188 jer_2016M
189 prop_binCRTT_muon2017_bin9
190 prop_binCRTT_electron2018_bin9
191 prop_binCRTT_muon2018_bin0
192 prop_binCROS_electron2016M_bin0
193 QCDScale_ZZtoLep
194 prop_binCRTT_muon2017_bin3
195 prop_binCROS_electron2018_bin4
196 prop_binCRTT_electron2018_bin3
197 prop_binCRTT_muon2017_bin10
198 prop_binCROS_muon2017_bin7_WrongSign
199 prop_binCROS_muon2016M_bin1
200 prop_binCROS_electron2018_bin9
201 mischarge_muon_2016M
202 TTest_muon_2017 $1.00^{+0.05}_{-0.05}$
203 prop_binCROS_muon2016M_bin10
204 prop_binCROS_electron2016M_bin3
205 prop_binCRTT_muon2018_bin6
206 jes_2018
207 prop_binCROS_muon2018_bin7
208 prop_binCRTT_muon2018_bin7
209 prop_binCROS_electron2016M_bin7
210 mischarge_muon_2017
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



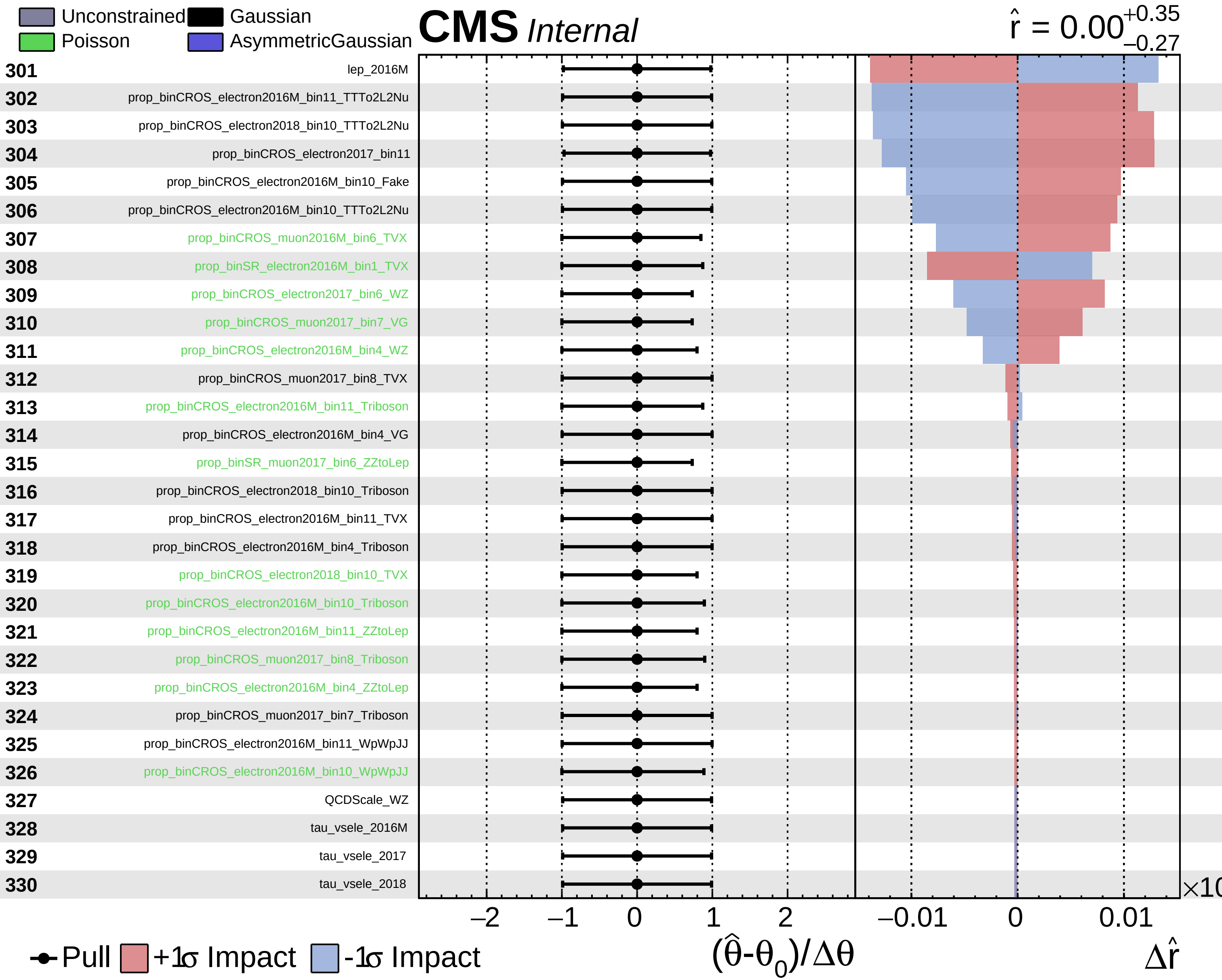
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.27}$
211 TTest_electron_2017 $1.00^{+0.06}_{-0.06}$
212 prop_binCROS_muon2016M_bin6_Fake
213 prop_binCRTT_muon2018_bin4
214 prop_binCRTT_muon2016M_bin1
215 prop_binCRTT_electron2016M_bin10
216 prop_binCROS_electron2016M_bin8
217 prop_binCRTT_electron2016M_bin6
218 TTest_electron_2016M $1.00^{+0.07}_{-0.07}$
219 prop_binCRTT_electron2016M_bin7
220 prop_binCROS_electron2017_bin10
221 prop_binCROS_electron2017_bin6_Fake
222 prop_binCRTT_muon2016M_bin4
223 prop_binCRTT_electron2018_bin8
224 prop_binCROS_muon2018_bin9
225 prop_binCROS_electron2017_bin9
226 prop_binCRF_electron2016M_bin0
227 mischarge_electron_2016M
228 prop_binCROS_electron2016M_bin5
229 prop_binCROS_electron2017_bin8
230 prop_binCROS_muon2017_bin9_WrongSign
231 prop_binCRTT_electron2017_bin7
232 prop_binCROS_electron2017_bin3
233 prop_binCRTT_electron2018_bin1
234 prop_binCRTT_electron2018_bin5
235 prop_binCROS_electron2018_bin7
236 prop_binCRTT_muon2017_bin6
237 prop_binCRTT_electron2016M_bin1
238 prop_binCRTT_muon2016M_bin11
239 prop_binCROS_electron2018_bin8
240 prop_binCRTT_muon2016M_bin8
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



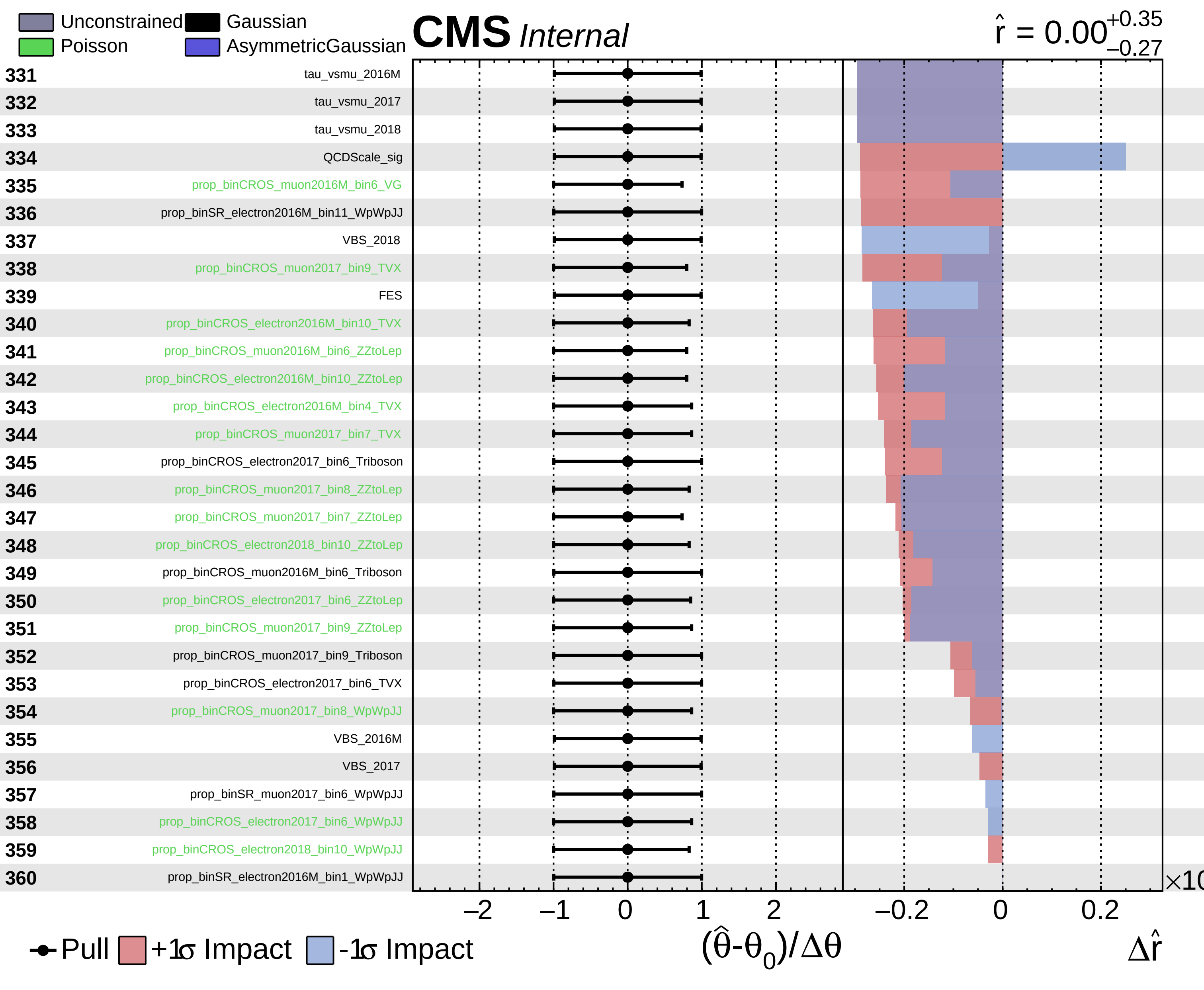
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.27}$
241 prop_binCRTT_muon2016M_bin10
242 prop_binCROS_muon2017_bin10
243 prop_binCROS_electron2017_bin7
244 prop_binCRTT_electron2017_bin11
245 TTest_muon_2018
$1.00^{+0.05}_{-0.05}$
246 prop_binCROS_muon2017_bin8_Fake
247 prop_binCROS_muon2016M_bin11
248 prop_binCRTT_electron2016M_bin2
249 mischarge_muon_2018
250 prop_binCRTT_electron2016M_bin11
251 prop_binCRF_electron2017_bin0
252 prop_binCRTT_electron2018_bin6
253 prop_binCRTT_muon2016M_bin3
254 prop_binCRTT_electron2018_bin7
255 prop_binCRTT_electron2016M_bin9
256 prop_binCRTT_electron2017_bin1
257 prop_binSR_muon2017_bin6_TTTo2L2Nu
258 prop_binCRTT_electron2016M_bin4
259 prop_binCROS_muon2017_bin11
260 puID
261 prop_binCRTT_electron2016M_bin0
262 prop_binCRTT_muon2016M_bin6
263 prop_binCRTT_muon2018_bin5
264 prop_binCRTT_electron2017_bin8
265 prop_binCROS_muon2017_bin7_Fake
266 prop_binCRTT_muon2017_bin1
267 prop_binCRTT_muon2016M_bin5
268 prop_binCRTT_muon2016M_bin9
269 prop_binCRTT_electron2018_bin4
270 prop_binCROS_muon2016M_bin9
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



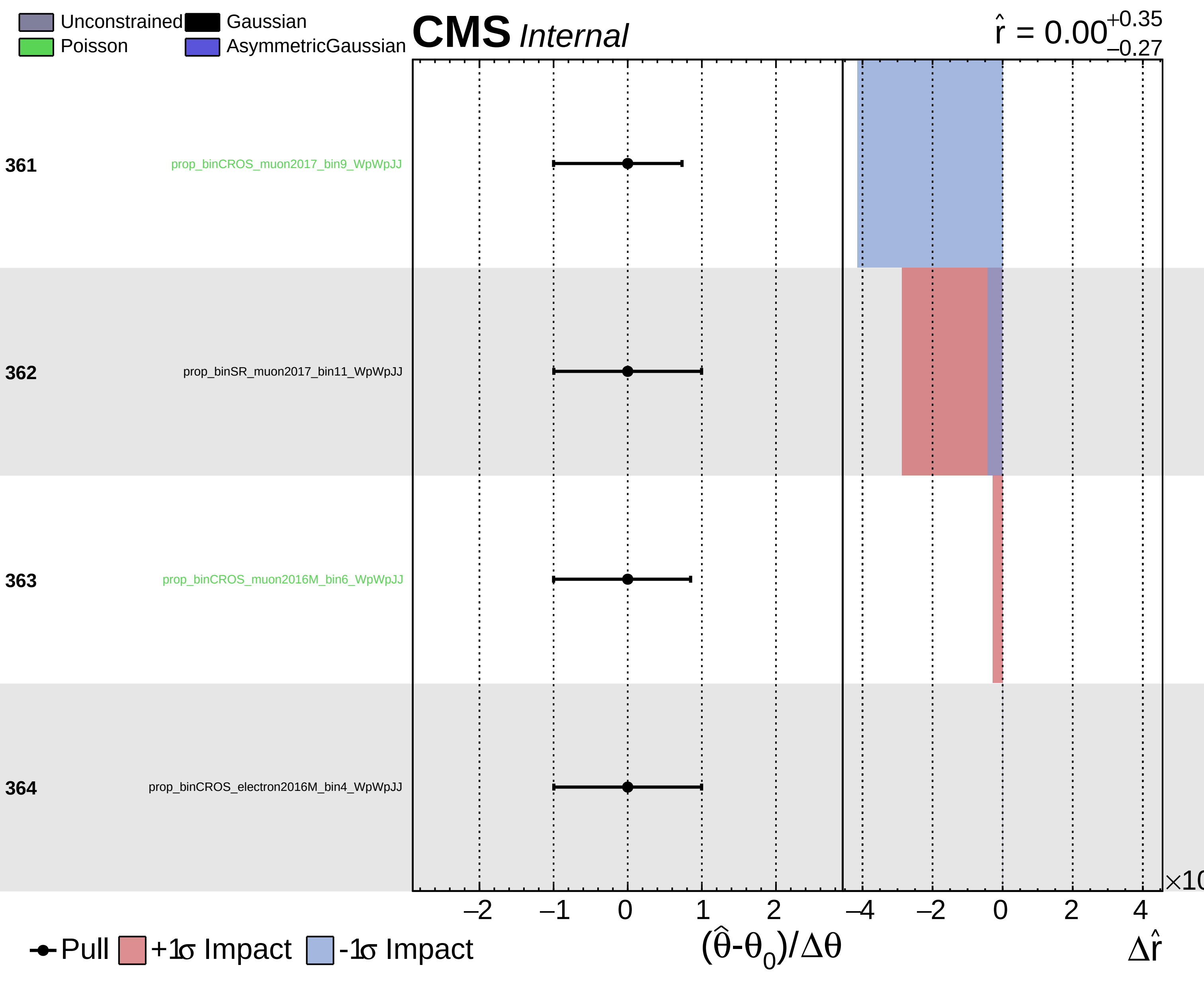
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.27}$
$1.00^{+0.05}_{-0.05}$
271 TTest_electron_2018
272 prop_binSR_electron2016M_bin1_WZ
273 prop_binCRTT_electron2016M_bin5
274 prop_binCROS_electron2016M_bin9
275 prop_binCROS_muon2018_bin11
276 prop_binSR_electron2016M_bin11_ZZtoLep
277 prop_binCROS_muon2016M_bin6_TTTo2L2Nu
278 prop_binCRTT_electron2017_bin10
279 mischarge_electron_2017
280 prop_binCROS_muon2017_bin8_TTTo2L2Nu
281 prop_binSR_electron2016M_bin1_TTTo2L2Nu
282 prop_binCROS_electron2018_bin11
283 lep_2018
284 prop_binCROS_muon2018_bin10
285 lep_2017
286 prop_binCROS_muon2017_bin9_TTTo2L2Nu
287 prop_binCROS_electron2016M_bin4_TTTo2L2Nu
288 prop_binSR_electron2016M_bin11_WrongSign
289 prop_binCROS_electron2016M_bin11_WrongSign
290 prop_binCROS_muon2017_bin9_Fake
291 prop_binSR_muon2017_bin6_TVX
292 prop_binCROS_electron2017_bin6_TTTo2L2Nu
293 prop_binSR_muon2017_bin6_Triboson
294 prop_binCROS_muon2017_bin7_TTTo2L2Nu
295 prop_binSR_electron2016M_bin1_ZZtoLep
296 mischarge_electron_2018
297 prop_binSR_electron2016M_bin1_Triboson
298 prop_binCRTT_muon2017_bin7
299 prop_binCROS_electron2018_bin10_Fake
300 prop_binCROS_electron2018_bin10_VG
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.27}$
301 lep_2016M
302 prop_binCROS_electron2016M_bin11_TTTo2L2Nu
303 prop_binCROS_electron2018_bin10_TTTo2L2Nu
304 prop_binCROS_electron2017_bin11
305 prop_binCROS_electron2016M_bin10_Fake
306 prop_binCROS_electron2016M_bin10_TTTo2L2Nu
307 prop_binCROS_muon2016M_bin6_TVX
308 prop_binSR_electron2016M_bin1_TVX
309 prop_binCROS_electron2017_bin6_WZ
310 prop_binCROS_muon2017_bin7_VG
311 prop_binCROS_electron2016M_bin4_WZ
312 prop_binCROS_muon2017_bin8_TVX
313 prop_binCROS_electron2016M_bin11_Triboson
314 prop_binCROS_electron2016M_bin4_VG
315 prop_binSR_muon2017_bin6_ZZtoLep
316 prop_binCROS_electron2018_bin10_Triboson
317 prop_binCROS_electron2016M_bin11_TVX
318 prop_binCROS_electron2016M_bin4_Triboson
319 prop_binCROS_electron2018_bin10_TVX
320 prop_binCROS_electron2016M_bin10_Triboson
321 prop_binCROS_electron2016M_bin11_ZZtoLep
322 prop_binCROS_muon2017_bin8_Triboson
323 prop_binCROS_electron2016M_bin4_ZZtoLep
324 prop_binCROS_muon2017_bin7_Triboson
325 prop_binCROS_electron2016M_bin11_WpWpJJ
326 prop_binCROS_electron2016M_bin10_WpWpJJ
327 QCDScale_WZ
328 tau_vsele_2016M
329 tau_vsele_2017
330 tau_vsele_2018
−2 −1 0 1 2
−0.01 0 0.01
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.35}_{-0.27}$
331 tau_vsmu_2016M
332 tau_vsmu_2017
333 tau_vsmu_2018
334 QCDScale_sig
335 prop_binCROS_muon2016M_bin6_VG
336 prop_binSR_electron2016M_bin11_WpWpJJ
337 VBS_2018
338 prop_binCROS_muon2017_bin9_TVX
339 FES
340 prop_binCROS_electron2016M_bin10_TVX
341 prop_binCROS_muon2016M_bin6_ZZtoLep
342 prop_binCROS_electron2016M_bin10_ZZtoLep
343 prop_binCROS_electron2016M_bin4_TVX
344 prop_binCROS_muon2017_bin7_TVX
345 prop_binCROS_electron2017_bin6_Triboson
346 prop_binCROS_muon2017_bin8_ZZtoLep
347 prop_binCROS_muon2017_bin7_ZZtoLep
348 prop_binCROS_electron2018_bin10_ZZtoLep
349 prop_binCROS_muon2016M_bin6_Triboson
350 prop_binCROS_electron2017_bin6_ZZtoLep
351 prop_binCROS_muon2017_bin9_ZZtoLep
352 prop_binCROS_muon2017_bin9_Triboson
353 prop_binCROS_electron2017_bin6_TVX
354 prop_binCROS_muon2017_bin8_WpWpJJ
355 VBS_2016M
356 VBS_2017
357 prop_binSR_muon2017_bin6_WpWpJJ
358 prop_binCROS_electron2017_bin6_WpWpJJ
359 prop_binCROS_electron2018_bin10_WpWpJJ
360 prop_binSR_electron2016M_bin1_WpWpJJ
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.35 −0.27
361
prop_binCROS_muon2017_bin9_WpWpJJ
362
prop_binSR_muon2017_bin11_WpWpJJ
363
prop_binCROS_muon2016M_bin6_WpWpJJ
364
prop_binCROS_electron2016M_bin4_WpWpJJ
−2 −1 0 1 2
−4 −2 0 2 4
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂